

SURFACE MOUNT TECHNOLOGY TRENDS IN INDIA

Use of SMT in electronics manufacturing is evolving in India. New services supporting SMT are coming up, helping to set new benchmarks and improve manufacturing

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Surface-mount technology (SMT) is one of the key factors for the shrinking electronics system-level packaging. The key drivers for SMT in India are the use of surface-mount devices (SMDs) in products developed indigenously by the Indian companies in the private as well as government sector, and the sourcing strategies of global electronics leaders and original equipment manufacturers (OEMs) to have their own manufacturing set-ups in India,

or outsourcing the manufacturing to Indian or global contract electronic manufacturers (CEMs).

With the electronics packaging being an ever-evolving field, and the focus on shrinking the size and having more packaging capability within the same or lesser real estate, component packaging has shrunk as well. Due to the condensed component structure, the component fan-out has increased multi-fold. To cater to this increased fan-out, printed circuit boards (PCBs) have moved from the conventional two-sided boards to multi-layer boards, with many manufacturers

choosing high-density interconnect (HDI) methods for their designs.

PCBs, components and systems

Though SMD components manufacturing in India is not in line with the global trends, some manufacturing is taking place. Majority of the components used for board manufacturing are still sourced from abroad.

However, interconnect structure and PCB fabrication are moving quite closely with the global trends. A few PCB fabrication houses are capable of fabricating advance-level boards using HDI techniques.

Most of the globally accepted low-loss laminate materials are available in India for the PCBs. Various metallic and organic surface finishes are also available. The cumulative result is advanced PCBs catering to both leaded and lead-free processes in India.

Areas of growth for PCB manufacturing include flexible PCBs, metal-clad PCBs and PCBs with embedded discretes. Flexible PCB manufacturing that has 2D flexural capability can be seen widely in India, though



Electronics assembly